



SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



COURSE CODE : CSA40

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COURSE NAME : MANAGEMENT INFORMATION SYSTEM

Reg NO: 192110204

4) Draw a UML diagram for ATM System using CASE tool. The banking system allows a customer to access the financial transactions by ATM System, it has a step-by-step process describe the work of this process and elaborate the what are the work can do by customer, banking system, administrator and technicians with the ATM system.

Aim:

To design a **UML Diagram** for an **ATM System** using a CASE (Computer-Aided Software Engineering) tool, illustrating the interactions between the **Customer, Banking System, Administrator, and Technicians** in performing financial transactions.

Procedure:

1. User Authentication

- The customer inserts the ATM card and enters the PIN for authentication.

2. Check Account Details

- The system verifies the PIN and retrieves the user's account information from the bank database.

3. Select Transaction Type

- The customer selects a transaction (withdrawal, deposit, balance inquiry, fund transfer, etc.).

4. Perform Transaction

- The banking system processes the requested operation (e.g., debits or credits the account).

5. Dispense Cash (If Withdrawal)

- If the transaction is a withdrawal, the ATM dispenses cash and updates the balance.

6. Print Receipt (Optional)

- The user can choose to print a receipt for transaction confirmation.

7. End Session

- The system logs out the user and ejects the ATM card after a period of inactivity or upon request.

8. Administrator Management

- The bank administrator manages customer accounts, sets transaction limits, and monitors suspicious activities.

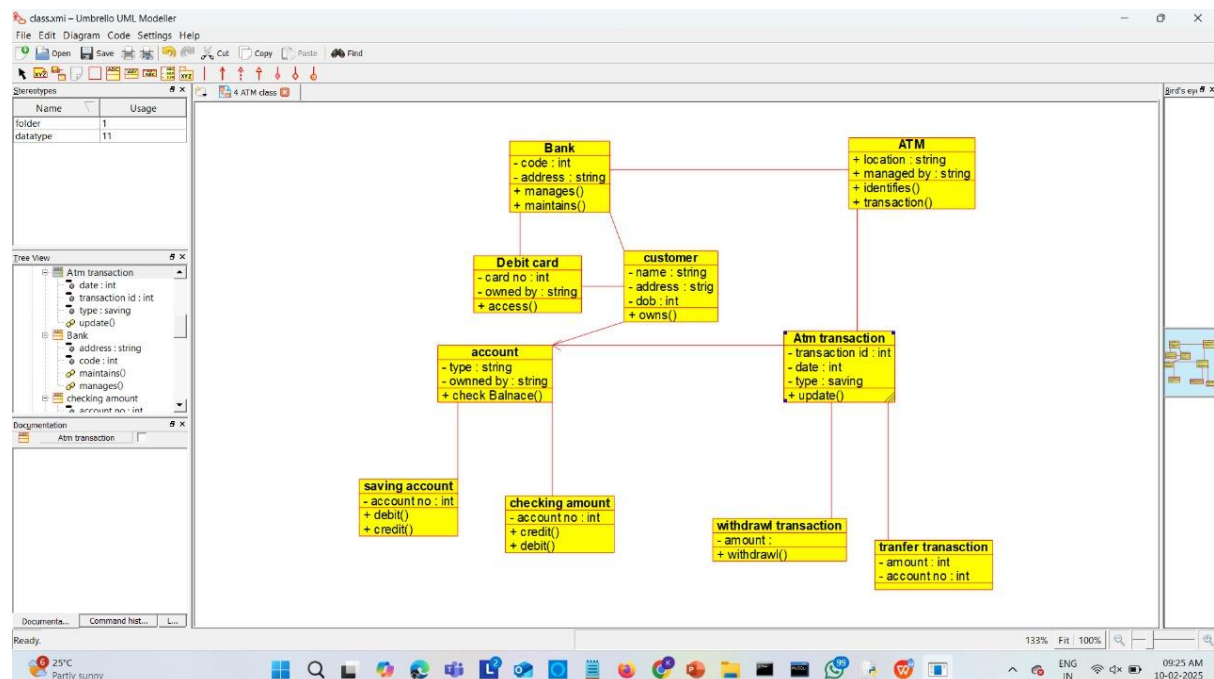
9. Technician Maintenance

- Technicians refill cash, repair hardware, and ensure the ATM system is functional.

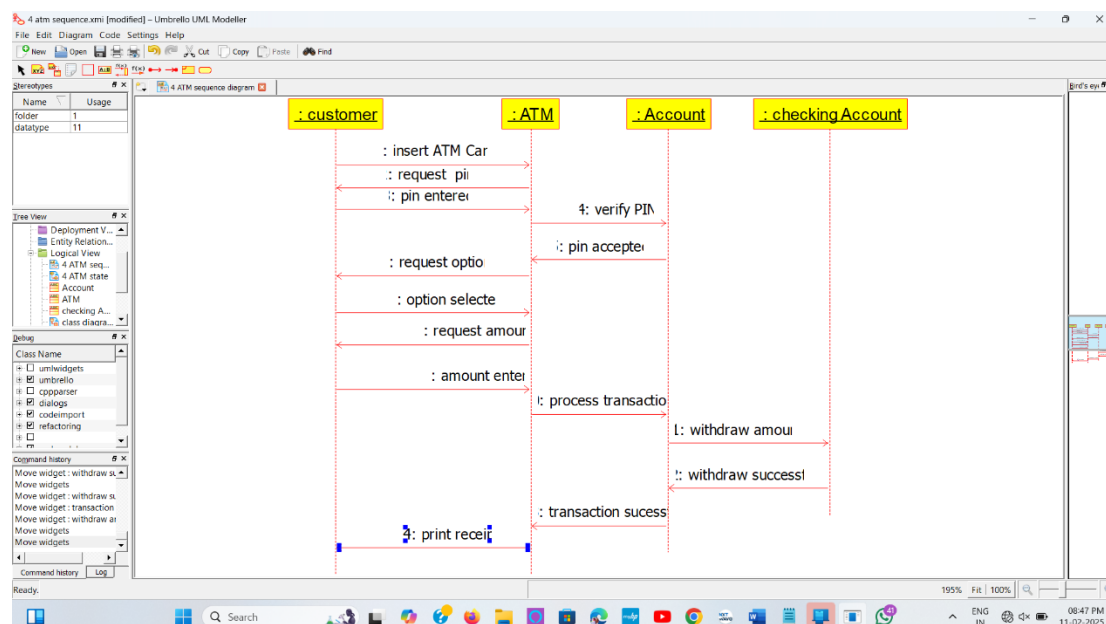
10. Bank System Updates

- The banking system updates transaction logs, account balances, and fraud detection reports.

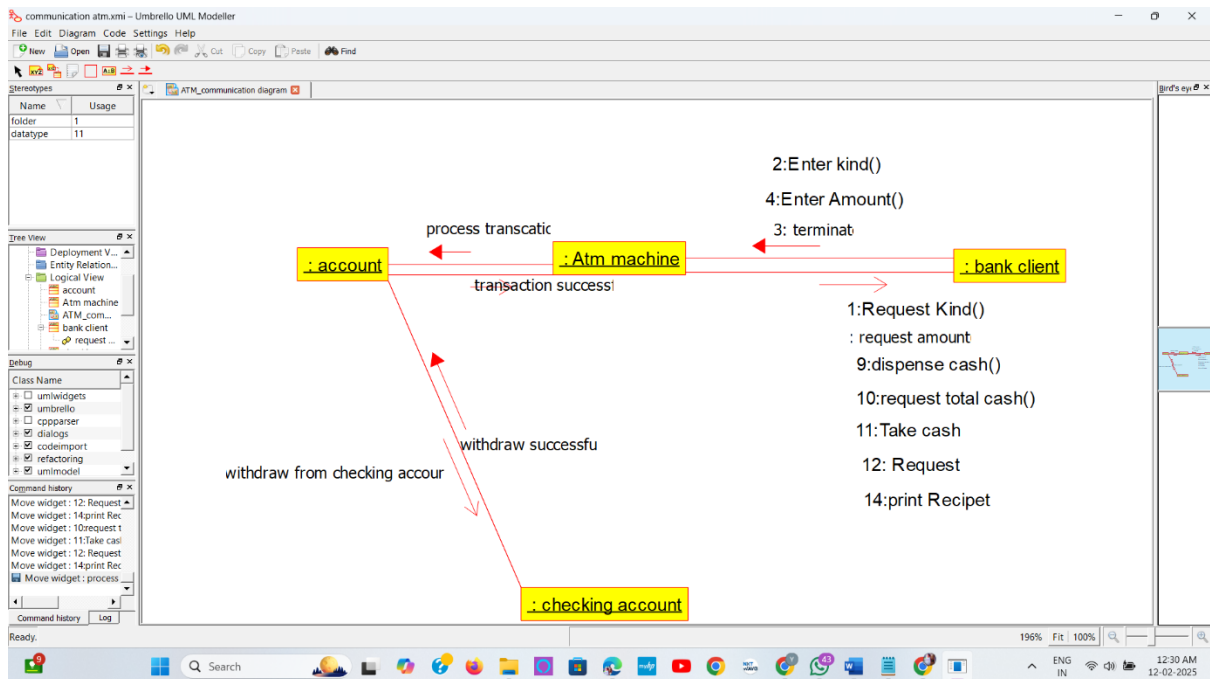
Class Diagram...



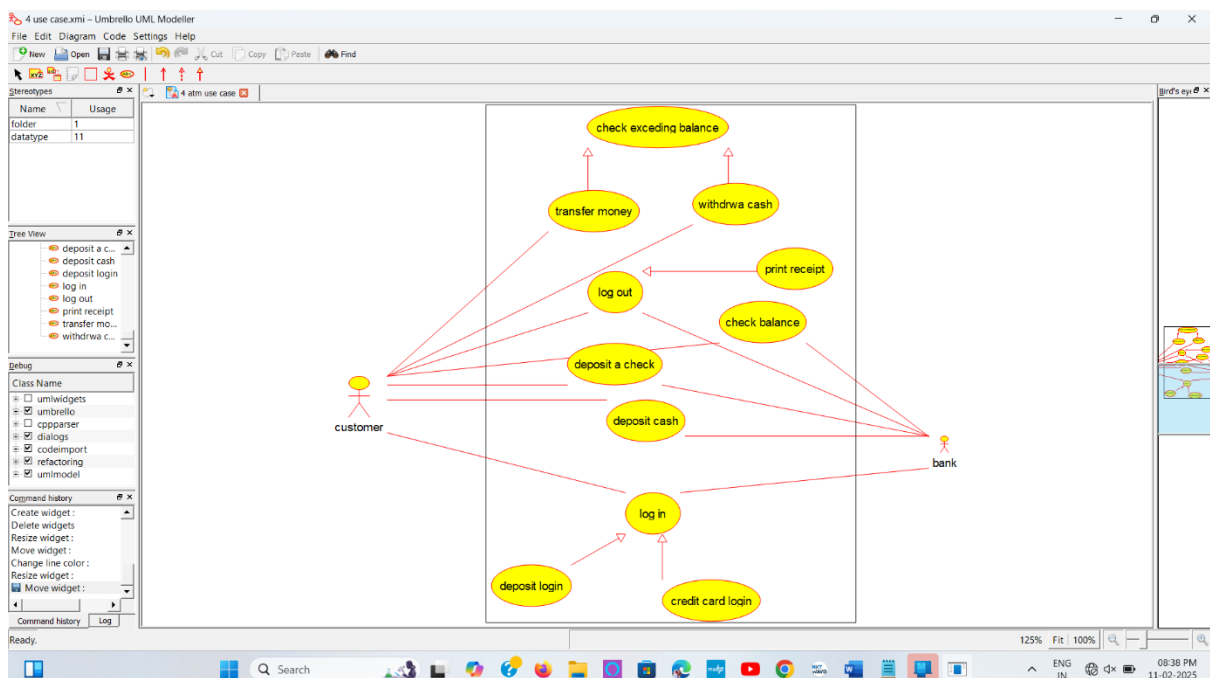
Sequence Diagram...



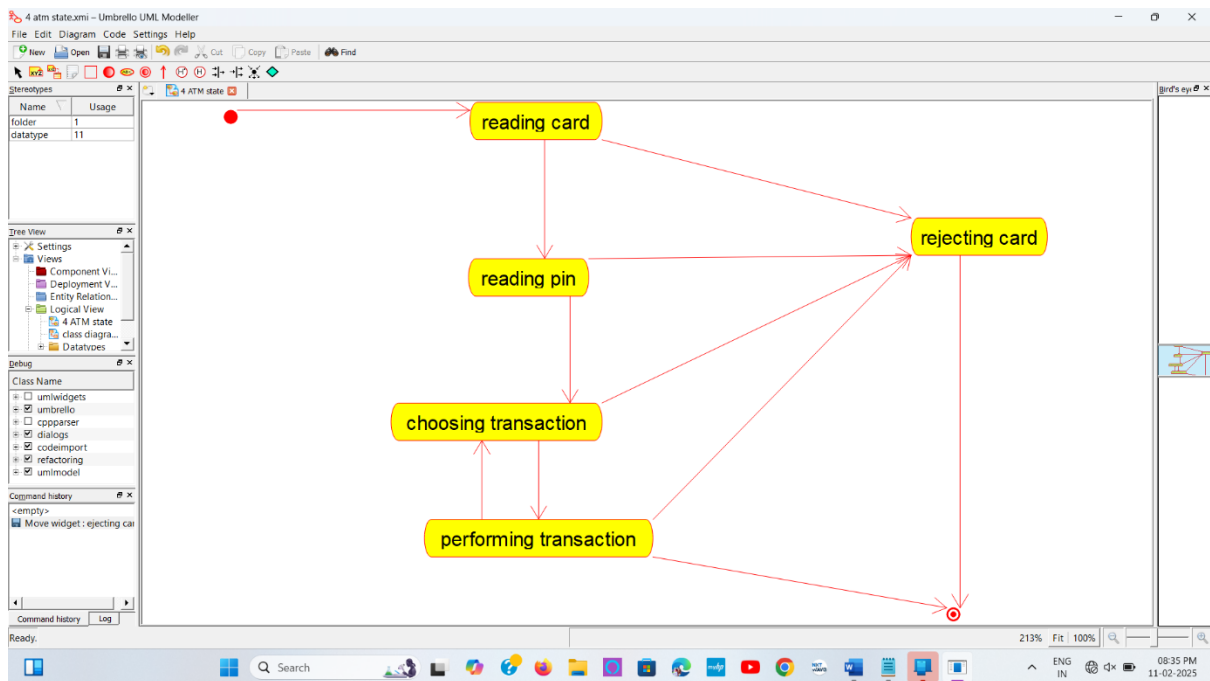
Communication Diagram...



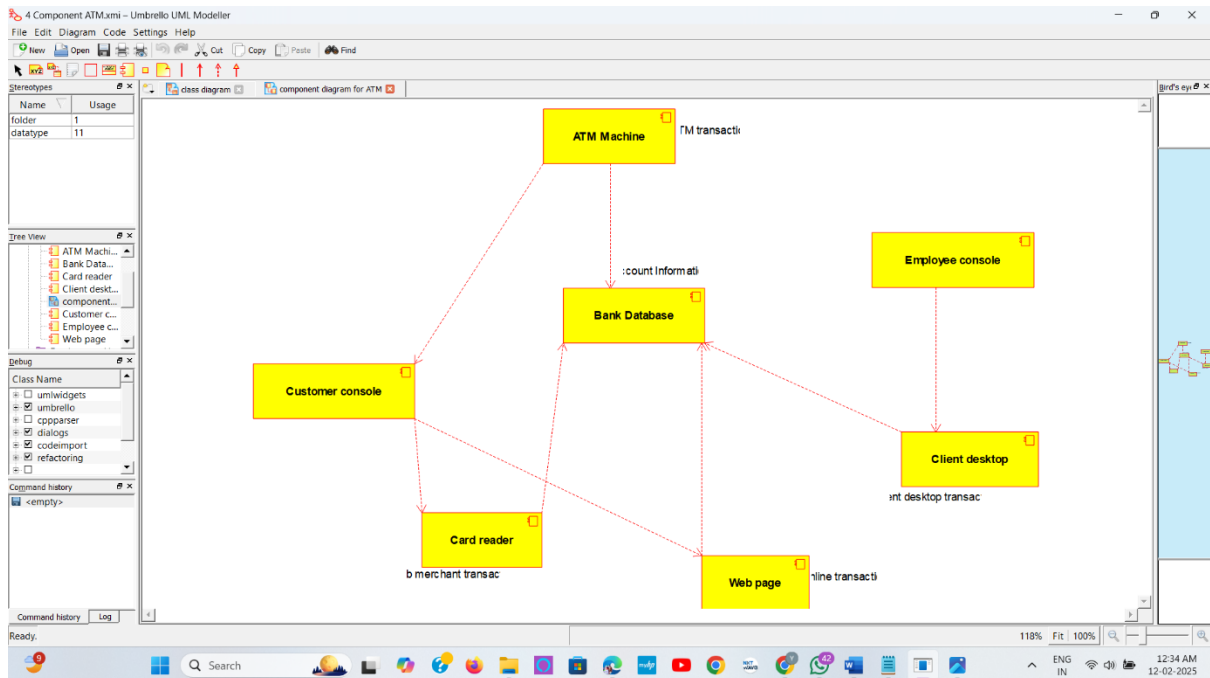
Use Case Diagram...



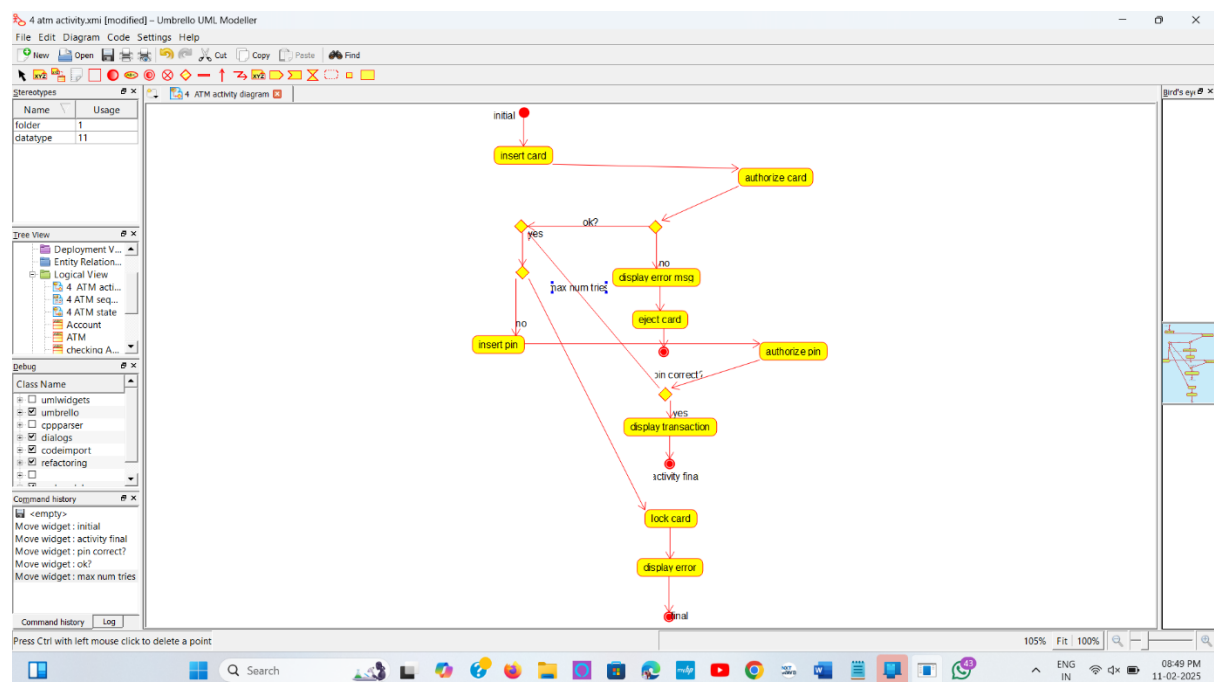
State Diagram...



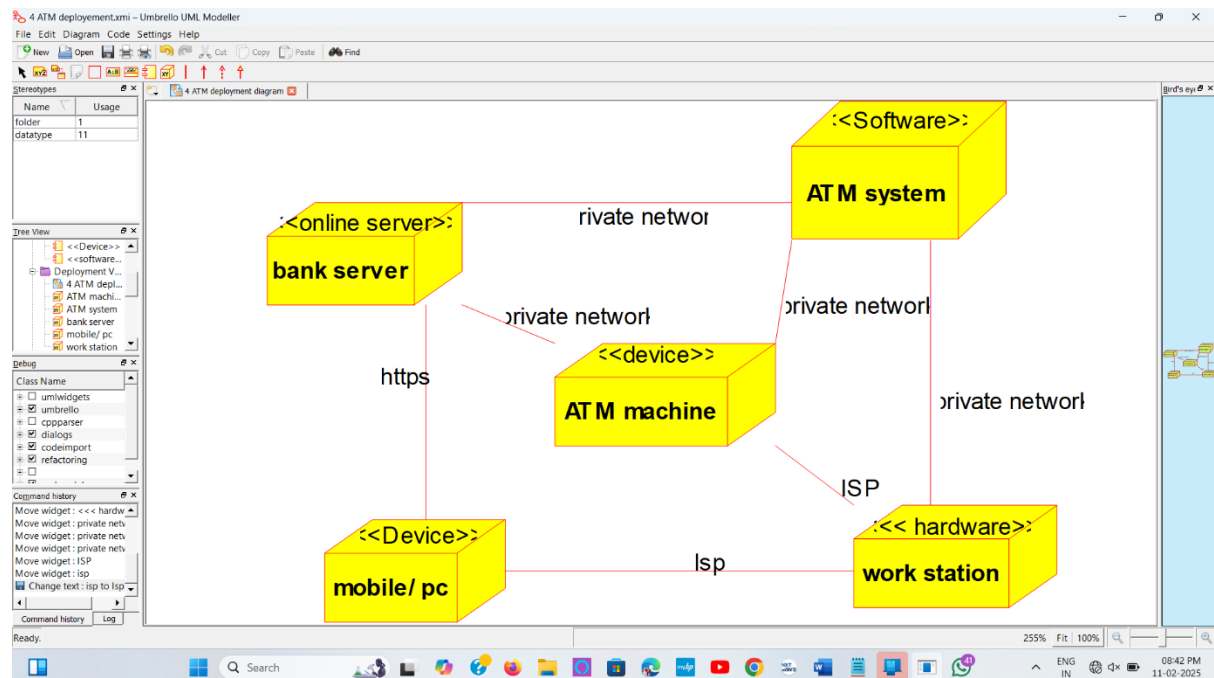
Component Diagram...



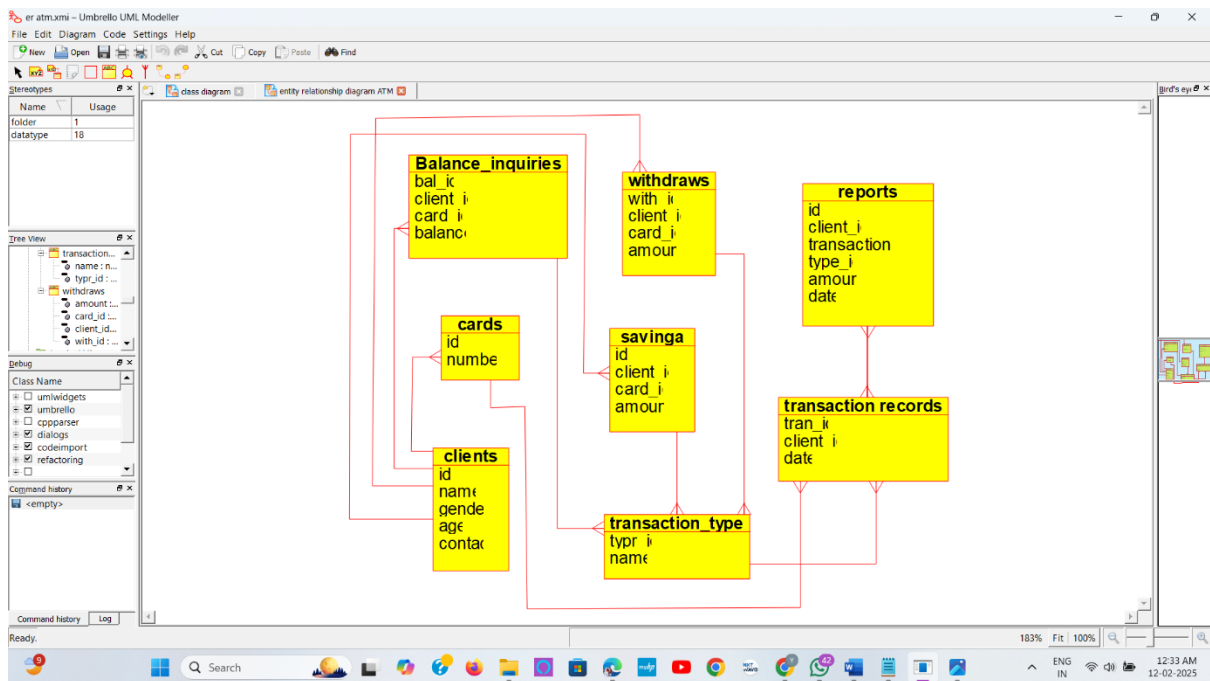
Activity Diagram...



Deployment Diagram...



Entity Relationship Diagram...



Result: The UML diagram for the ATM System was successfully designed using a CASE tool, showing interactions between the Customer, Banking System, Administrator, and Technicians. It illustrates the step-by-step process of authentication, transactions, and maintenance. This helps in better system implementation, security, and optimization for banking